

Figure 7: Compression of Apache when one core is enabled

File systems

In the file system, it is important to know how to handle the file and its allocation within the memory. In the file system, the memory is fully responsible for resource management, its usage, processing and retrieval whenever the file is needed by the user. Without the file management, the hardware is simply a bunch of circuits with no knowledge of the file's location, its starting address and ending address.

Different types of file systems like Ext2, Ext3, Ext4, ReiserFS, FAT and NTFS have various performance levels.

Figure 8 shows the read/write process on the respective file systems. In this graph, Ext versions perform differently on the same hardware. In this figure, it can be seen that the reading/writing speed changes based on the Ext version. Reading and writing speed is measured in Mb/second.

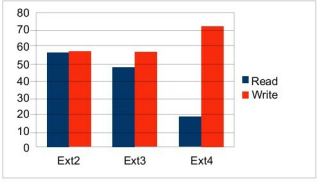


Figure 8: File system performance chart

Boot time

Boot speed matters a lot in the case of operating systems, and problems often arise during booting. The application that one chooses to be configured on the system start-up dictates the boot speed of Linux based systems.

Some applications are not useful for the average user. Often, these applications reduce the boot time of the system. Set these applications on a manual mode, which means they will not be starting up at the boot time. The default boot time of the system is approximately one minute, which is shown in Figure 9. If you need these applications, start them manually from the command line. The unnecessary applications are:

- Bluetooth
- Cups and Cups-browser
- Console
- irqbalance
- cron, etc

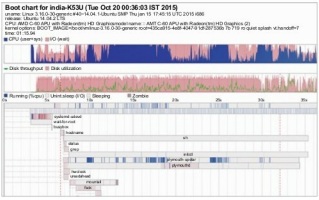


Figure 9: Default OS boot processes

The correct way to disable an upstart service is as follows:

```
echo "manual" >> /etc/init/myjob.conf
```

Now, the upstart services will not start automatically.

When the unnecessary applications mentioned above were closed, the boot time recorded was approximately 45 seconds, which is shown in Figure 10.

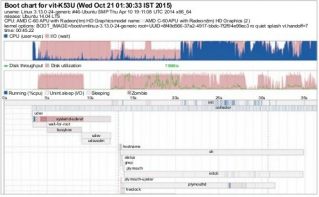


Figure 10: Modified boot processes

Kernel recompilation/optimisation

The kernel provided by all distros is probably the best one for almost everybody. The Linux kernel is compiled in many different ways. Recompiling the kernel can decrease kernel size and its overall behaviour. Changing certain parameters in the source code might also yield some change in the system's performance. Recompile the Linux kernel with different options, install it and then boot the system from it.

Kernel recompilation steps: Download the kernel from the official site, as follows:

```
linux-4.2.1.tar
cd /usr/src/
tar xf /path/to/linux-4.2.1.tar
```

Clear the old object files or configuration